

Flume mooring specification — OSU Large Wave Flume (HWRL), 1:50 test articles

CONSERVATIVE SPECIFICATION, 2026-09-25. It was issued ahead of full verification: Phase D (the

full FloatSim case matrix) follows before the test (see *Follow-ups*). Built by `build_spec.py`

from FloatSim results (`spec_statics.json`, `spec_extremes.json`, `spec_extreme_buoy.json`,

`line_hardware.json`). Basis and derivations: `DESIGN-BASIS.md`.

Page 1 — Facility assumptions to CONFIRM WITH HWRL BEFORE BUYING HARDWARE

None of these is a confirmed facility fact. The design rests on all of them.

#	Assumption	Value this specification uses	If it proves false
F1	Anchors can be fixed on the flume walls 0.72 m above still water (the pin plane)	anchors at $x = \pm 5.0$ m, $y = \pm 1.83$ m, $z = +0.717$ m	The lines slope into the splash zone and the attachment must drop towards the waterline, where the pretension tilts the pinned buoys (calm-water tilt 4.9° cluster, 9.1° platform). Redesign.
F2	Wall anchor rating	per-anchor working-load limits in the tables below: 311 N per platform anchor (cluster 80 N, buoy 20 N)	Lower pretension or softer lines, with more offset. Redesign.
F3	Wave height per period, steepness	$H/\lambda \leq 0.08$ at 2.7 m depth: H 0.5 m only for $T \geq 2.01$ s (and from 2.35 s by the 3° tilt rule), H 0.35 m for $T \geq 1.675$ s	Cases outside the wavemaker envelope drop out.
F4	Water depth	2.7 m	The drift bound falls for deeper water; periods unchanged.
F5	Test-section position vs the wavemaker and beach	a clean regular-wave window $\geq 2\text{--}3$ min at the article, which sits on the flume centreline	Shorter windows do not settle the lightly damped resonances ($\zeta \approx 3\text{--}4\%$).
F6	Tracking field of view	surge $-0.3 \dots +1.0$ m (down-flume), sway ± 0.1 m, plus the article footprint (platform 2.06 m wide at 45°)	The extreme cases must stop earlier (see *Test procedure*).
F7	Standard soft-mooring hardware	elastic shock cord: stiffness and elongation per the tables below (secant stiffness within -13% / $+30\%$ of target over the working stretch), dry weight ≈ 0.3 N/m	Without it, the lines need farther anchors (longer lines, lower strain) or another soft element. Redesign.

Headline

	single buoy	cluster	4×4 platform
lines (legs)	4	4	8
attachment	spar-top collar $r = 0.2$ m, $+0.717$ m	each moored spar's pin, $+0.717$ m	each moored spar's pin, $+0.717$ m
cord stiffness k (band)	2.01 N/m (1.75–2.61)	8.30 N/m (7.22–10.79)	16.64 N/m (14.47–21.63)
pretension T_0 per line/leg	4.00 N	14.42 N	28.85 N
unstretched length L_0	3.133 m	3.213 m	2.489 / 2.659 m
at-rest stretch	2.01 m	1.74 m	1.73 m
peak stretch (worst case)	3.28 m	3.23 m	3.14 m
elongation capacity $\times 1.25$	4.10 m (131 % of L_0)	4.04 m (126 % of L_0)	3.92 m (158 % of L_0)
max line tension / working load $\times 3$	6.6 / 20 N	26.8 / 80 N	52.2 / 157 N
anchor working-load limit $\times 3$	20 N	80 N	311 N
min tension / T_0 (≥ 0.15)	0.42	0.19	0.21
surge / sway / yaw period	16.4 / 22.1 / 0.86 s	16.1 / 22.6 / 8.81 s	16.0 / 21.6 / 10.17 s
pull stiffness surge / sway	7.4 / 3.7 N/m	31.1 / 13.7 N/m	126.2 / 60.5 N/m
pull stiffness yaw	3.34 N·m/rad	24.22 N·m/rad	312.54 N·m/rad

Frame for all coordinates (the TRUE flume frame): origin at the article's calm centre on the flume

centreline at still water; x down-flume (the wave direction), y across, z up. Anchors and

attachments are at $z = +0.717$ m, the pin plane.

What is conservative in these numbers:

- the **drift sum**: the full recorded fixed-body drift bound, applied in-run at each spar's

waterline, plus FloatSim's own mean wave force on the moving body;

- +60 % pretension** on the cluster and platform, over the Phase C design (line stiffness k

kept);

- working loads $\times 3$** and **elongation capacity $\times 1.25$** on the worst-case FloatSim values;
- the worst cases are $H = 0.5$ m at $T = 2.35$ s (the largest admissible drift) and 2.65 s (the

tilt/heave resonance).

1. Layout and line make-up (all articles)

- An X-spread of four soft lines to the flume walls, 20.1° off the flume axis. The platform's four lines are 2-leg bridles, 8 legs, to the two pins of each up- and down-stream half-row.

- Each line or leg is an **elastic shock cord**, with short stiff leads/shackles if needed, of unstretched length L_0 . It is pre-stretched to the pretension T_0 when attached: tension at rest $\approx k \times (\text{chord} - L_0)$.

- Attachments are at the pin plane (+0.717 m):

- **single buoy**: a stiff radial collar (crossbar) of radius $r = 0.2 \text{ m}$ on the spar top, one line at the point facing each anchor. Yaw period 0.86 s, below the wave band. It holds heading for tracking;

- **cluster**: each spar's pin;

- **platform**: the pins of the 8 up- and down-stream row spars.

- Pretension is set by L_0 . Check it at installation against the at-rest tension and stretch in the tables.

2. Per-article line tables

The **up-flume lines (anchors at x = -5 m) carry the drift** and set every peak value. The

down-flume lines relax towards T0 and below. **Buy and rig every line to its article's

up-flume values** (the headline) so the lines are interchangeable.

2a. Single buoy

line	anchor (x, y, z) m	attachment (x, y, z) m	L0 unstretched	k target (band -13 / +30 %)	pretension T0	at-rest stretch	peak stretch	elongation capacity ×1.25	max tension	working load ×3
1 (buoy)	(-5.000, +1.830, +0.717)	(-0.188, +0.069, +0.707)	3.133 m	2.01 N/m (1.75–2.61)	4.00 N	2.01 m	3.28 m	4.10 m (131 % of L0)	6.6 N	20 N
2 (buoy)	(-5.000, -1.830, +0.717)	(-0.188, -0.069, +0.707)	3.133 m	2.01 N/m (1.75–2.61)	4.00 N	2.01 m	3.28 m	4.10 m (131 % of L0)	6.6 N	20 N
3 (buoy)	(+5.000, +1.830, +0.717)	(+0.188, +0.069, +0.707)	3.133 m	2.01 N/m (1.75–2.61)	4.00 N	2.01 m	2.03 m	2.54 m (81 % of L0)	4.1 N	12 N
4 (buoy)	(+5.000, -1.830, +0.717)	(+0.188, -0.069, +0.707)	3.133 m	2.01 N/m (1.75–2.61)	4.00 N	2.01 m	2.03 m	2.54 m (81 % of L0)	4.1 N	12 N

anchor (x, y, z) m	lines	anchor design load (sum of line maxima)	anchor working-load limit ×3
(-5.000, +1.830, +0.717)	1	6.6 N	20 N
(-5.000, -1.830, +0.717)	2	6.6 N	20 N
(+5.000, +1.830, +0.717)	3	4.1 N	12 N
(+5.000, -1.830, +0.717)	4	4.1 N	12 N

2b. Cluster (4 buoys at 45°)

line	anchor (x, y, z) m	attachment (x, y, z) m	L0 unstretched	k target (band -13 / +30 %)	pretension T0	at-rest stretch	peak stretch	elongation capacity ×1.25	max tension	working load ×3
1 (buoy2)	(-5.000, +1.830, +0.717)	(-0.295, +0.295, +0.714)	3.213 m	8.30 N/m (7.22–10.79)	14.42 N	1.74 m	3.23 m	4.04 m (126 % of L0)	26.8 N	80 N
2 (buoy3)	(-5.000, -1.830, +0.717)	(-0.295, -0.295, +0.714)	3.213 m	8.30 N/m (7.22–10.79)	14.42 N	1.74 m	3.23 m	4.04 m (126 % of L0)	26.8 N	80 N
3 (buoy1)	(+5.000, +1.830, +0.717)	(+0.295, +0.295, +0.714)	3.213 m	8.30 N/m (7.22–10.79)	14.42 N	1.74 m	1.74 m	2.17 m (68 % of L0)	14.4 N	43 N
4 (buoy4)	(+5.000, -1.830, +0.717)	(+0.295, -0.295, +0.714)	3.213 m	8.30 N/m (7.22–10.79)	14.42 N	1.74 m	1.74 m	2.17 m (68 % of L0)	14.4 N	43 N

anchor (x, y, z) m	lines	anchor design load (sum of line maxima)	anchor working-load limit ×3
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(-5.000, +1.830, +0.717)	1	26.8 N	80 N
(-5.000, -1.830, +0.717)	2	26.8 N	80 N
(+5.000, +1.830, +0.717)	3	14.4 N	43 N
(+5.000, -1.830, +0.717)	4	14.4 N	43 N

2c. 4×4 platform (16 buoys at 45°)

line	anchor (x, y, z) m	attachment (x, y, z) m	L0 unstretched	k target (band – 13 / +30 %)	pretension T0	at-rest stretch	peak stretch	elongation capacity ×1.25	max tension	working load ×3
1 (buoy6)	(-5.000, +1.830, +0.717)	(-0.884, +0.884, +0.715)	2.489 m	16.64 N/m (14.47–21.63)	28.85 N	1.73 m	3.14 m	3.92 m (158 % of L0)	52.2 N	157 N
2 (buoy7)	(-5.000, +1.830, +0.717)	(-0.884, +0.295, +0.715)	2.659 m	16.64 N/m (14.47–21.63)	28.85 N	1.73 m	3.10 m	3.87 m (146 % of L0)	51.5 N	155 N
3 (buoy10)	(-5.000, -1.830, +0.717)	(-0.884, -0.295, +0.715)	2.659 m	16.64 N/m (14.47–21.63)	28.85 N	1.73 m	3.10 m	3.87 m (146 % of L0)	51.5 N	155 N
4 (buoy11)	(-5.000, -1.830, +0.717)	(-0.884, -0.884, +0.715)	2.489 m	16.64 N/m (14.47–21.63)	28.85 N	1.73 m	3.14 m	3.92 m (158 % of L0)	52.2 N	157 N
5 (buoy1)	(+5.000, +1.830, +0.717)	(+0.884, +0.884, +0.715)	2.489 m	16.64 N/m (14.47–21.63)	28.85 N	1.73 m	1.70 m	2.12 m (85 % of L0)	28.3 N	85 N
6 (buoy4)	(+5.000, +1.830, +0.717)	(+0.884, +0.295, +0.715)	2.659 m	16.64 N/m (14.47–21.63)	28.85 N	1.73 m	1.70 m	2.13 m (80 % of L0)	28.3 N	85 N
7 (buoy13)	(+5.000, -1.830, +0.717)	(+0.884, -0.295, +0.715)	2.659 m	16.64 N/m (14.47–21.63)	28.85 N	1.73 m	1.70 m	2.13 m (80 % of L0)	28.3 N	85 N
8 (buoy16)	(+5.000, -1.830, +0.717)	(+0.884, -0.884, +0.715)	2.489 m	16.64 N/m (14.47–21.63)	28.85 N	1.73 m	1.70 m	2.12 m (85 % of L0)	28.3 N	85 N

anchor (x, y, z) m	lines	anchor design load (sum of line maxima)	anchor working-load limit ×3
(-5.000, +1.830, +0.717)	1, 2	103.8 N	311 N
(-5.000, -1.830, +0.717)	3, 4	103.8 N	311 N
(+5.000, +1.830, +0.717)	5, 6	56.6 N	170 N
(+5.000, -1.830, +0.717)	7, 8	56.6 N	170 N

Anchor working-load limits exceed the 250 N flag only if the platform row says so. **F2 must be confirmed for those values.**

3. Hardware

- **Elastic shock cord: recommended for all three articles.**

- It must hold its secant stiffness within -13% / $+30\%$ of the target over the working stretch

(at-rest to peak).

- Below -13% , the extreme offset leaves the field of view. Above $+30\%$, the surge period falls to ~ 14 s and the slack margin vanishes.

- Required elongation capacity: see the headline ($\times 1.25$ on the FloatSim peak stretch). If the cord is shorter than L0 (a stiff lead in series), its strain rises in proportion.

- Its hysteresis is not modelled. It damps the slow modes and shifts the mean position by half the loading/unloading gap, which the pull test measures.

- **Pulley + counterweight (constant tension): rejected.** All its restoring is geometric, about $20\times$ softer in surge (surge periods 75–108 s). Under the drift it moves the articles past the $+1.0$ m tracking limit from $H = 0.12$ m (cluster, platform) or 0.2 m (buoy), and 3–5 m in the extreme band. FloatSim cannot model it; the screen is in closed form (`line_hardware.py`).

4. Tank validation targets (installed mooring, before any waves)

Check the installed mooring against these FloatSim predictions:

- the static pull (a horizontal pull through the article centre at the pin plane, force against offset);
- free decays from ~0.3 m surge, ~0.1 m sway, ~5° yaw.

The buoy's yaw decay is not predictable in FloatSim; its yaw period comes from the collar stiffness and the buoy's yaw inertia.

	surge period	sway period	yaw period	surge pull K / F at 0.25, 0.5 m	sway pull K / F at 0.1 m	yaw pull K / M at 5°
buoy	16.4 s	22.1 s	0.86 s	7.4 N/m / 1.9, 3.7 N	3.7 N/m / 0.37 N	3.34 N·m/rad / 0.291 N·m
cluster	16.1 s	22.6 s	8.81 s	31.1 N/m / 7.8, 15.5 N	13.7 N/m / 1.37 N	24.22 N·m/rad / 2.112 N·m
platform	16.0 s	21.6 s	10.17 s	126.2 N/m / 31.5, 63.0 N	60.5 N/m / 6.05 N	312.54 N·m/rad / 27.267 N·m

The surge and sway periods include the coupling to the buoys' tilt. Decay damping (FloatSim, round 1) is $\zeta \approx 8 \rightarrow 4$ % per cycle in surge and $5 \rightarrow 3$ % in sway.

5. Predicted envelope

Pretension against slack (worst cases, FloatSim, drift sum; pass: $T_{\min} \geq 0.15 T_0$):

article	T0 scale	T0 per line/leg	T = 2.35 s: T_min/T0, surge max	T = 2.65 s: T_min/T0, surge max
cluster	×1	9.05 N	0.09 FAIL , 1.39 m	0.10 FAIL , 0.95 m
cluster	×1.2	10.84 N	0.09 FAIL , 1.33 m	0.11 FAIL , 0.90 m
cluster	×1.4	12.63 N	0.11 FAIL , 1.30 m	0.17 PASS, 0.88 m
cluster	×1.6	14.43 N	0.19 PASS, 1.29 m	0.26 PASS, 0.87 m
platform	×1	18.04 N	0.04 FAIL , 1.29 m	0.04 FAIL , 0.98 m
platform	×1.2	21.65 N	0.04 FAIL , 1.25 m	0.05 FAIL , 0.94 m
platform	×1.4	25.25 N	0.10 FAIL , 1.23 m	0.12 FAIL , 0.92 m
platform	×1.6	28.86 N	0.21 PASS, 1.23 m	0.23 PASS, 0.92 m

Worst cases at the specified pretension (FloatSim, the conservative drift sum; buoy at

$T_0 = 4.0$ N on the collar; cluster and platform at +60 % T_0):

article	case	min / max line tension	T_min / T0 (≥ 0.15)	mean offset (range)	max tilt	min line clearance to local surface	validity
buoy	H 0.5 m, T 2.65 s, drift sum, T_0 4.04 N	1.7 / 6.6 N	0.42 PASS	0.71 m (0.57–0.86)	29.4°	+0.21 m	indicative (tilt > 5.7°)
cluster	H 0.5 m, T 2.35 s, drift sum, T_0 14.43 N	2.7 / 26.8 N	0.19 PASS	1.17 m (1.03–1.29) > +1.0 m	22.3°	+0.33 m	indicative (tilt > 5.7°)
cluster	H 0.5 m, T 2.65 s, drift sum, T_0 14.43 N	3.8 / 25.6 N	0.26 PASS	0.76 m (0.64–0.87)	29.8°	+0.33 m	indicative (tilt > 5.7°)
platform	H 0.5 m, T 2.35 s, drift sum, T_0 28.86 N	6.1 / 52.2 N	0.21 PASS	1.09 m (0.95–1.23) > +1.0 m	20.3°	+0.36 m	indicative (tilt > 5.7°)
platform	H 0.5 m, T 2.65 s, drift sum, T_0 28.86 N	6.6 / 51.5 N	0.23 PASS	0.78 m (0.63–0.92)	28.7°	+0.37 m	indicative (tilt > 5.7°)

- **Operational band ($H \leq 0.12$ m).**

- Mean offset $\leq \sim 0.03$ m at the drift bound (static), and line tensions stay within ± 10 % of T_0

statically.

- At the tilt/pitch resonance the buoys tilt **9.2–9.6° at $H = 0.04$ m** and ****~16–17° at**

$H = 0.12$ m** (buoy, cluster; FloatSim sweeps).

- **Small-angle validity:** FloatSim here is LEVEL1 (small angles, valid to 0.1 rad = 5.7°). Every

prediction near the pitch/tilt resonance, line loads included, is **indicative across the whole operational band**, and all the worst cases above are beyond it. LEVEL2 is required for resonant predictions (follow-up).

- **Not predicted:** the single buoy's extremes near its moored pitch resonance ($H = 0.5$ m at $T = 2.35$ s; untested at 2.55 s). FloatSim diverges there at 28° pitch. No prediction until LEVEL2; see *Test procedure*.

6. Test procedure notes

- **Before waves:** measure each line's at-rest tension (or stretch) against the tables; the pull stiffness and decay periods against §4.
- **Extreme cases in increasing H** (0.2 → 0.35 → 0.5 m) at each period.
 - Monitor offset live. Stop a case if the down-flume offset approaches the tracking limit (e.g. 0.9 m of the +1.0 m), or a line nears slack or its working load.
 - Do NOT stiffen the lines to fix the field of view.
- **Dropped buoy extremes** (no FloatSim prediction until LEVEL2): run them LAST, with a height ramp (0.2 → 0.35 → 0.5 m). Watch roll, yaw and line tension; stop on any growth of roll or yaw.
 - Heading 0 (waves along the flume axis) for every article. The articles are mirror-symmetric about it, so yaw is not wave-driven; a misalignment would drive it.
 - Skip cases with $H/\lambda > 0.08$, and H = 0.5 m below $T = 2.35$ s.

7. Follow-ups before the test (deferred today)

Item	What it could change
BEM regeneration at NT = 36 (flume databases' 12-sided waterline; ~1.7 h platform BEM)	All heave/tilt resonances ~0.04–0.08 s shorter; the fine period band moves. Envelope loads change little (the peak height is damping-controlled).
Full Phase D (442 cases)	The criteria across the whole matrix, incl. operational-band dynamic tensions and offsets, and the criterion-6 interference check. Could flag cases, or revise T0 or anchor loads.
LEVEL2 and the resonant-subset rerun	Resonant tilts, line loads and offsets, all indicative today; the buoy's dropped extremes get a prediction. Could raise or lower the peak loads.
Drift mechanism	The sum used today adds FloatSim's own mean force (moving-body submerged drag, 20–70 % of the bound at H = 0.5 m) to the recorded fixed-body splash-zone bound. The mechanisms are distinct (FloatSim has no splash zone), so this is not the same force twice; it is conservative because the bound assumes a fixed body. A moving-body splash-zone model could lower the offsets and tensions.
Static-solver false convergence (tracker)	None for these results: every static solve is residual-checked.
HWRL confirmations F1–F7	Any of them can force a redesign (page 1).